Transport Layer (L4)

Services:

- End-to-end byte stream transmission that is transparent to applications .

Connection type:

- Connection- based (connection-oriented - TCP)
- Connectionless (connectionless - UDP)

TCP/UDP header format

TCP Segment Header Format

Bit #	0	7	8	15	16	23	24	31
0	Source Port				Destination Port			
32	Sequence Number							
64	Acknowledgment Number							
96	Data Offset	Res	Flags		Window Size			
128	Header and Data Checksum				Urgent Pointer			
160...	Options							

UDP Datagram Header Format

Bit #	0	7	8	15	16	23	24	31
0	Source Port				Destination Port			
32	Length				Header and Data Checksum			

Port numbers

The port number length is 16 bits: 0-65535

TCP port 80 != UDP port 80 (both protocols have their own 16-bit identifier range)

Identifying a connection:
Protocol + IP $_A$ + PORT $_A$ + IP $_B$ + PORT $_B$

List of well-known and registered services:
/ etc / services

Let's list the contents of the file and find the default port ID of some well-known services:
 # less / etc / services

FTP, Telnet, SSH, SMTP, POP3, IMAP, DNS, HTTP, HTTPs

Port numbers

Port number range	Port number block
0 - 1023	Well-known port numbers
1024 - 49151	Registered port numbers
49152 - 65535	Private and/or dynamic port numbers

TCP-based communication

Transmission characteristics:

- Connection- based
- Reliable
- Orderly

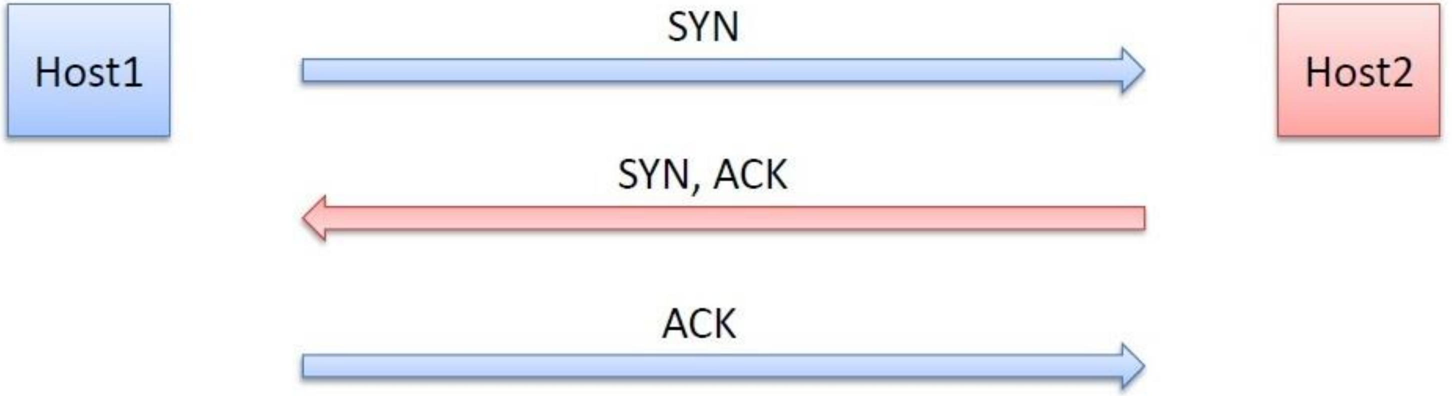
Connection-based connection:

- Relationship building phase
- Data transfer
- Disconnection phase

TCP connection setup

3 lépéses kézfogás algoritmus (3-way handshake)

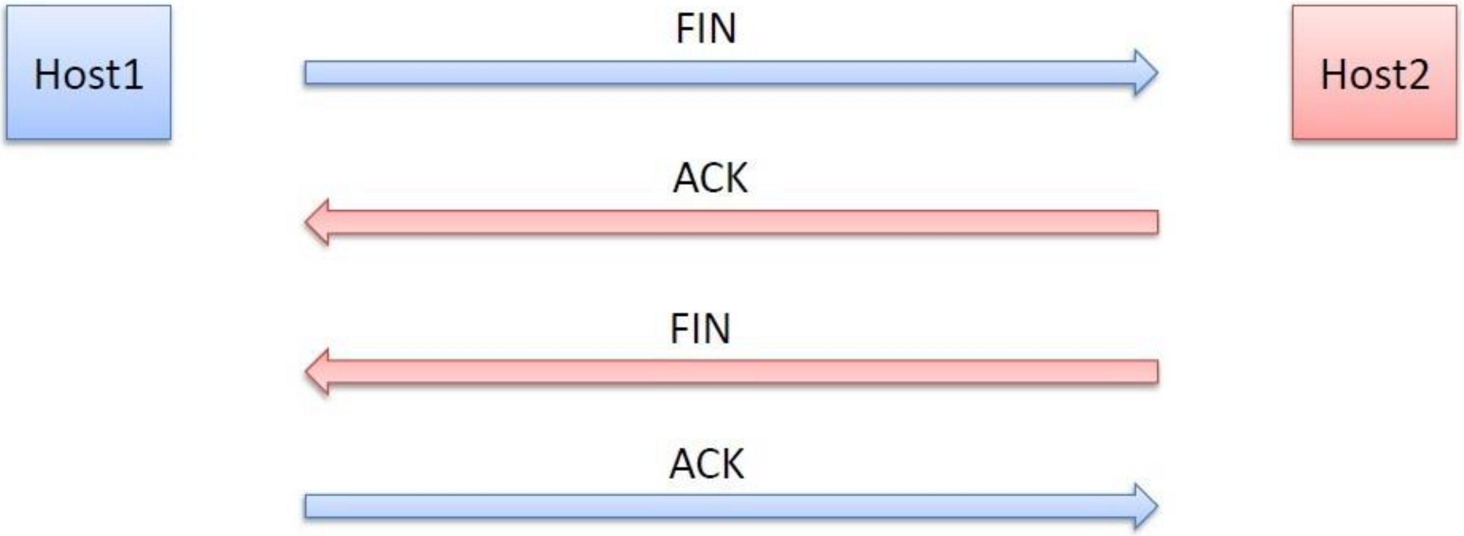
1. SYN (synchronize)
2. SYN, ACK (acknowledgement)
3. ACK



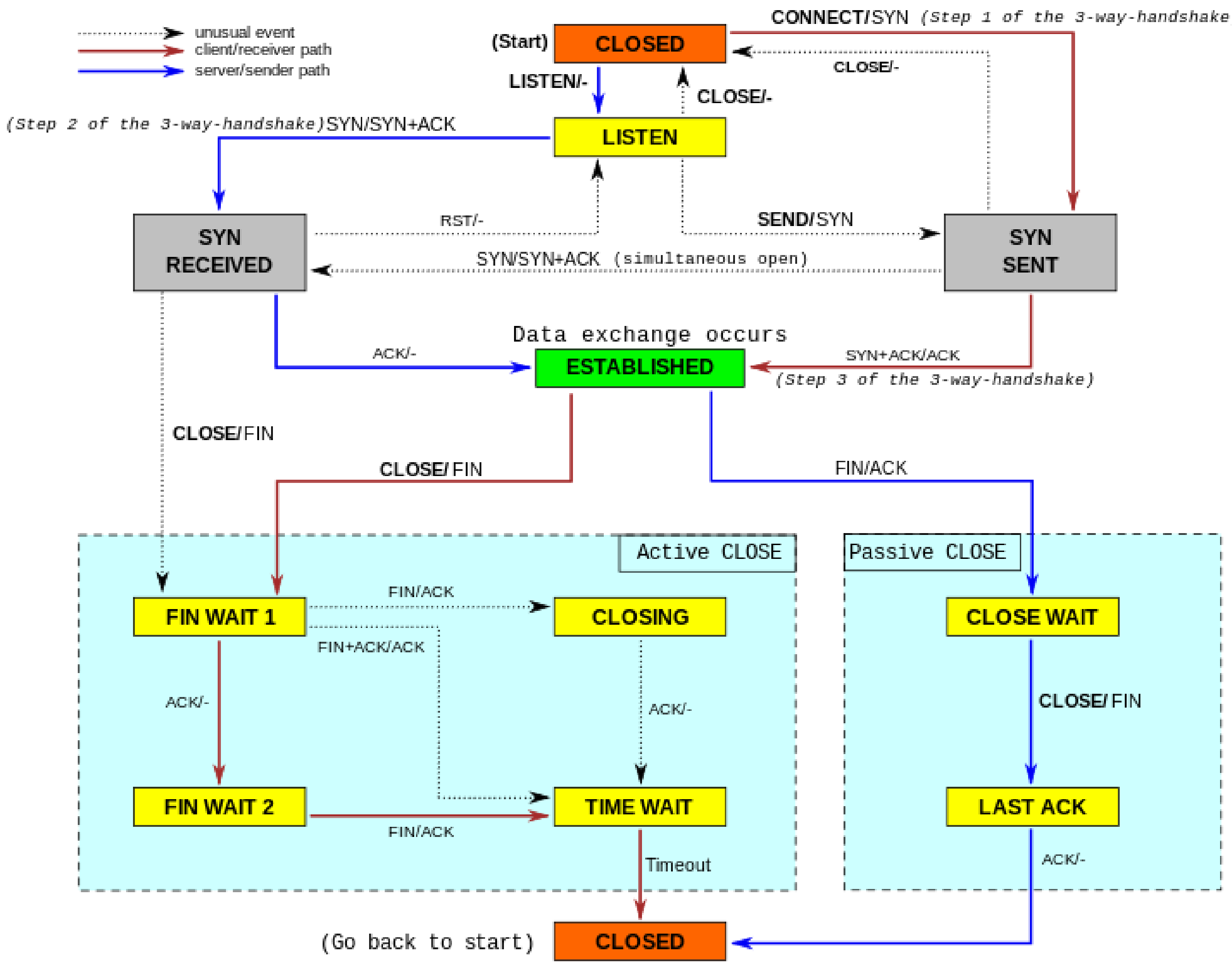
TCP connection termination

4 lépéses kapcsolatbontás (4-way handshake)

1. FIN
2. ACK
3. FIN
4. ACK



Finite state machine handling TCP connection



Source: wikipedia.org

TCP connection manager finite state machine states

Status	Description
CLOSED	No active or pending connections
LISTEN	The server is waiting for a call to arrive.
SYN RCVD	Connection request received, waiting
SYN SENT	initiated
ESTABLISHED	Normal data transmission status
FIN WAIT 1	The application announced that it was done with its tasks.
FIN WAIT 2	The other party has agreed to disconnect the connection.
TIME WAIT	Wait until all packages die out
CLOSING	Both parties tried to break the connection simultaneously
CLOSE WAIT	The other party initiated the demolition
LAST ACK	Wait until all packages die out

Querying network connections

To query your computer's current network connections:

 # netstat

To list the output of a command with paging:

 # netstat | more

More important command line switches:

-a : list all connections regardless of connection status (listening also lists sockets)
-c : refresh output every second
-n : list the IP addresses and port IDs of the nodes in numerical form
-p : list both the PID and name of the program associated with the connection
-r : list the machine's routing table = *route* command
-s : protocol level statistics
-t : list TCP port connections
-u : list UDP port connections
-v : more verbose output

Queryong network connections

Task:

Let's issue the following commands and then interpret the outputs in detail!

 # netstat -anp | more

 # netstat - at

 # netstat - oh

 # netstat - from

 # netstat - saw

 # netstat -lu

 # netstat -s

Task

Determine if we have an active communication web connection with a machine with an IP number N.N.N.N!

```
netstat –na | grep N.N.N.N
```

netcat (nc)

Chat

PC1: 192.168.0.107   PC2: 192.168.0.108:1234

PC2: **`nc -l -p 1234`**

PC1: **`nc 192.168.0.108 1234`**

nc

File transfer (from PC1 to PC2)

PC1: 192.168.0.107 PC2: 192.168.0.108:1234

PC2: **`nc –v –w 30 -p 1234 –l > output.txt`**

PC1: **`nc –v –w 2 192.168.0.108 1234 < input`**

nc

TCP port scanning

PC1: 192.168.0.107 PC2: 192.168.0.108:1234

PC2: **`nc 192.168.0.108 –p 9999 -l`**

PC1: **`nc –zv 192.168.0.108 9000-10000`**

nmap

TCP port scanning

PC1: 192.168.0.107 PC2: 192.168.0.108:1234

PC2: `nc 192.168.0.108 –p 9999 -l`

PC1: `nmap 192.168.0.108`

Task

Check if the service provider web server is accessible on TCP port 80 on the machine with IP number 195.228.245.1 !

```
nc –v 195.228.245.1 80
```

```
nmap 195.228.245.1 –p 80
```